UNIVERSIDAD DE GUAYAQUIL

FACULTAD DE CIENCIAS, MATEMATICAS Y FISICAS

CARRERA DE SOFTWARE

MATERIA:

GESTION DE LA CONFIGURACION DEL SOFTWARE

TEMA DE ACTIVIDAD:

CI/CD COMPLETO CON JENKINS

DOCENTE:

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ESTUDIANTE:

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CICLO ESCOLAR:

2025 – 2026 CICLO II

URL GITHUB:

https://github.com/Cinthia03/cicd-web-demo.git

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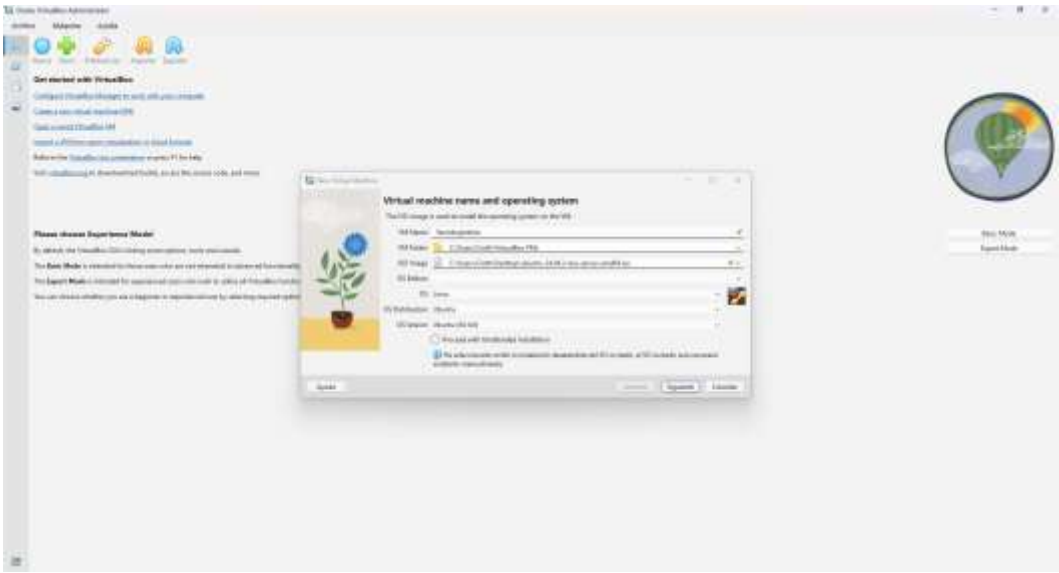
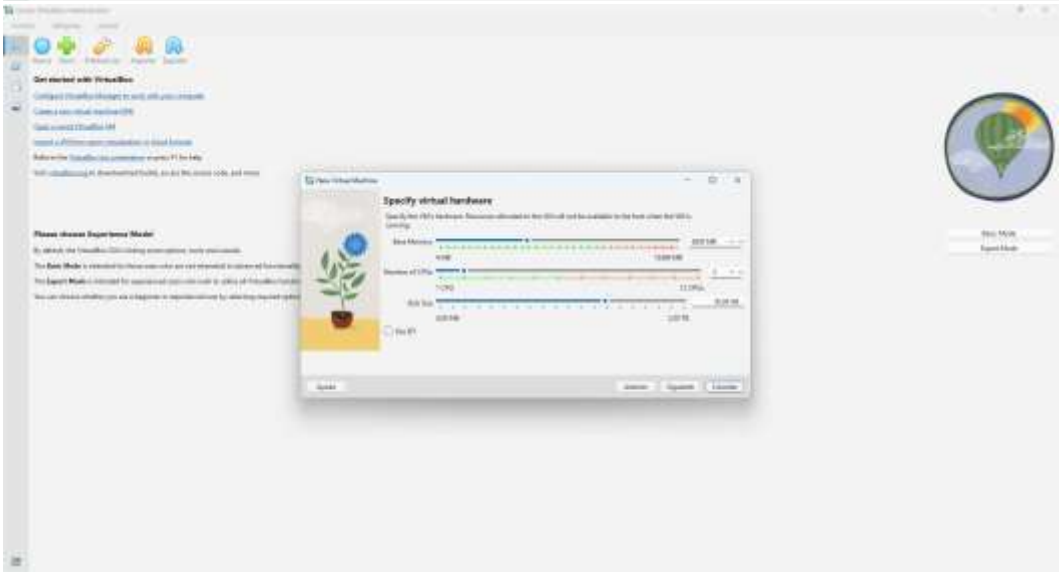
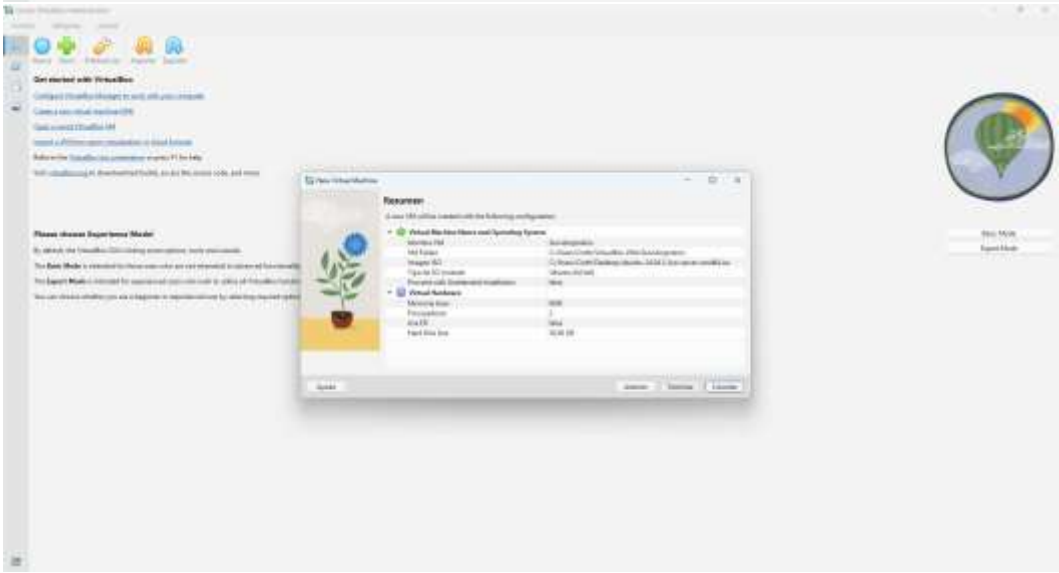
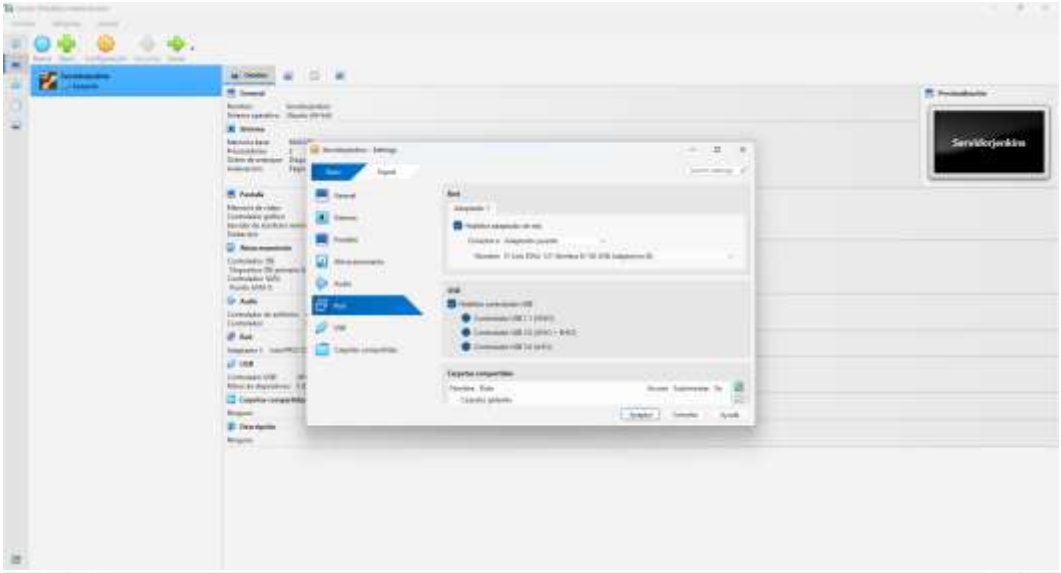
JENKINS

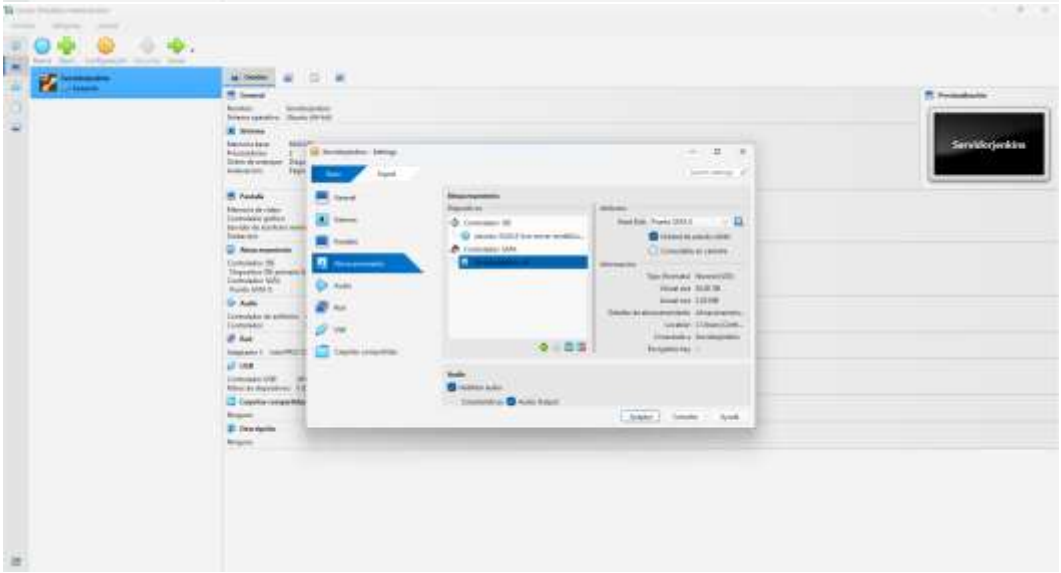
1. REQUISITOS DEL ENTORNO

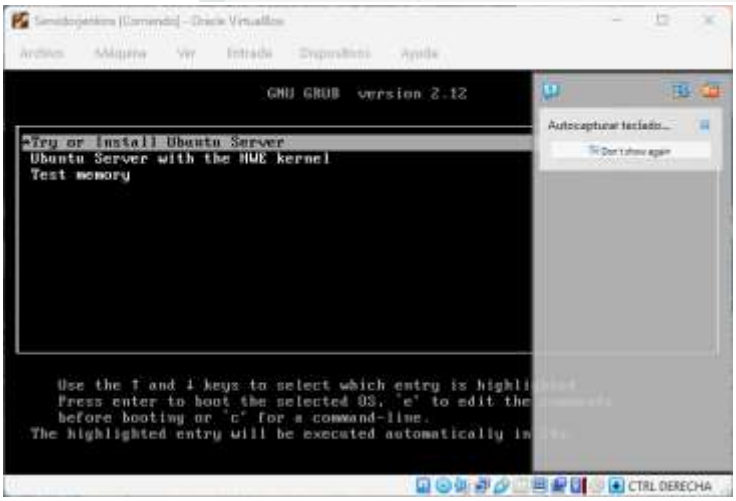


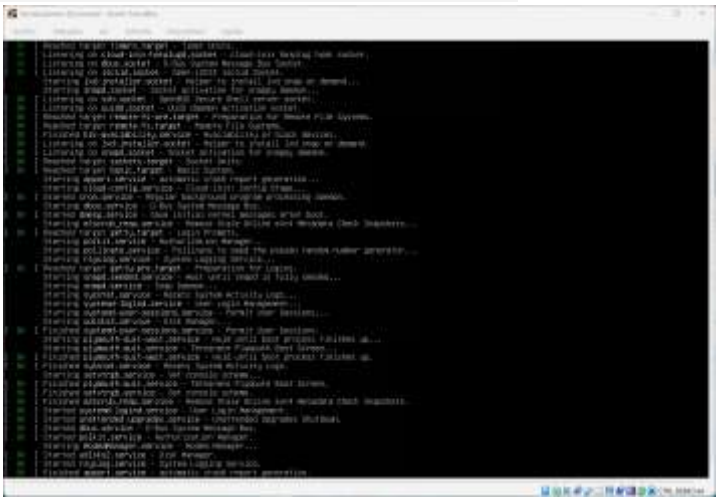


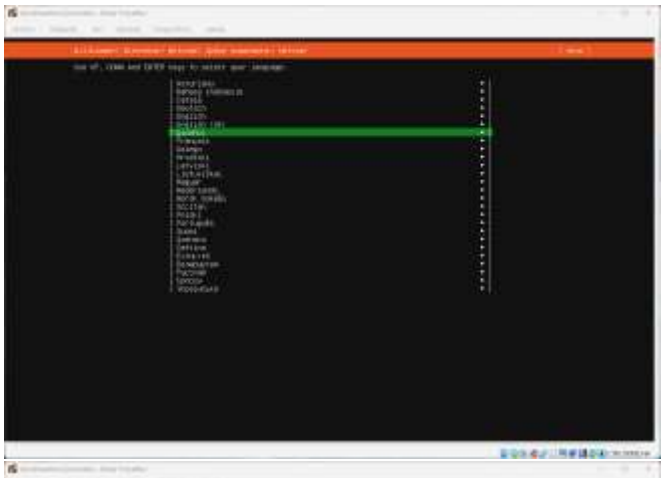


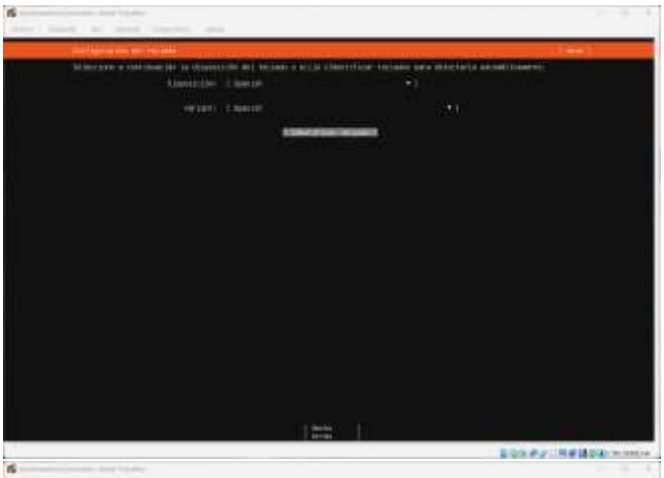


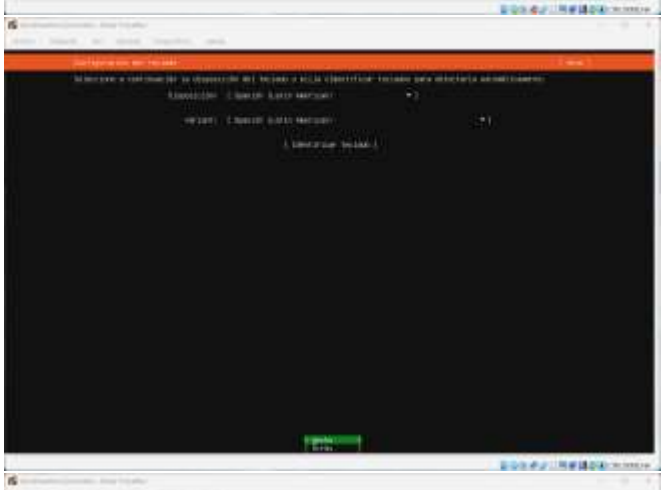


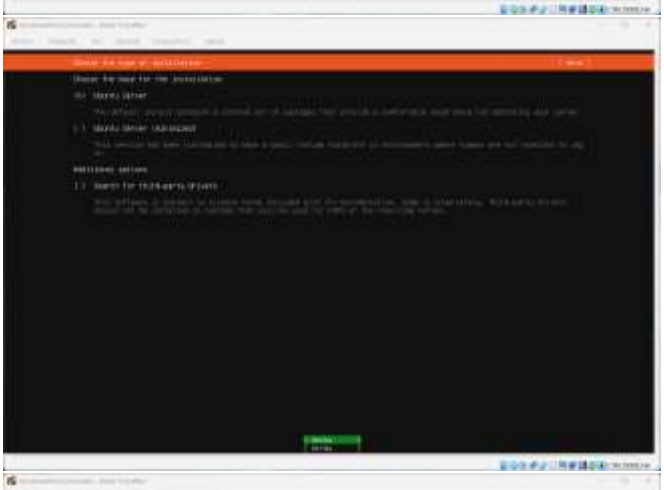

GNU GRUB version 2.12
Try or Install Ubuntu Server
Ubuntu Server with the HWE kernel
Test memory
Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line.
The highlighted entry will be executed automatically in 30s.
Autocapturar teclado...
Don't show again
CTRL DERECHA

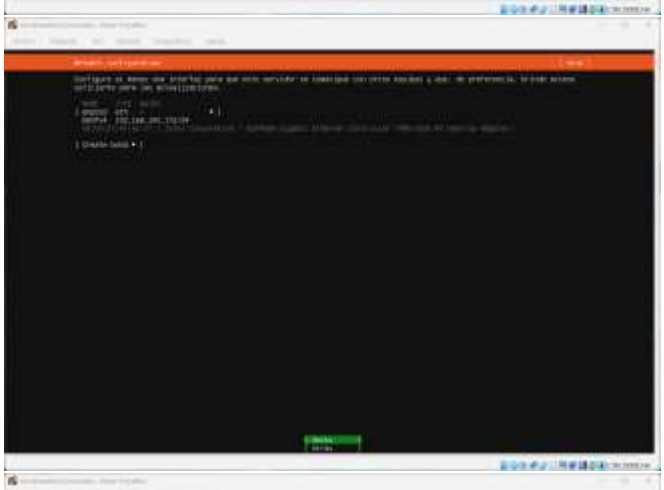


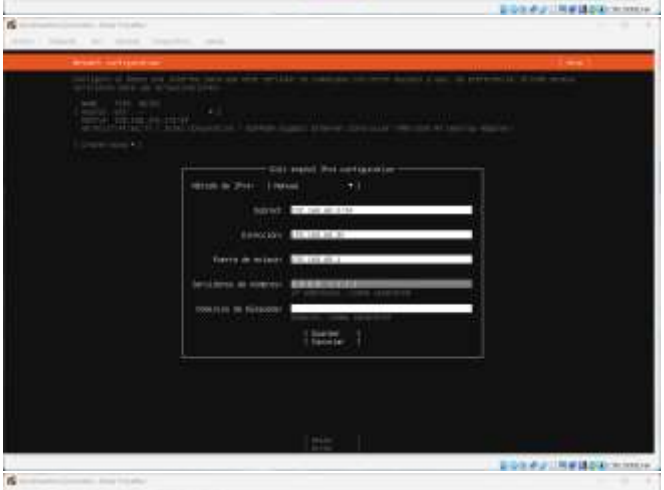


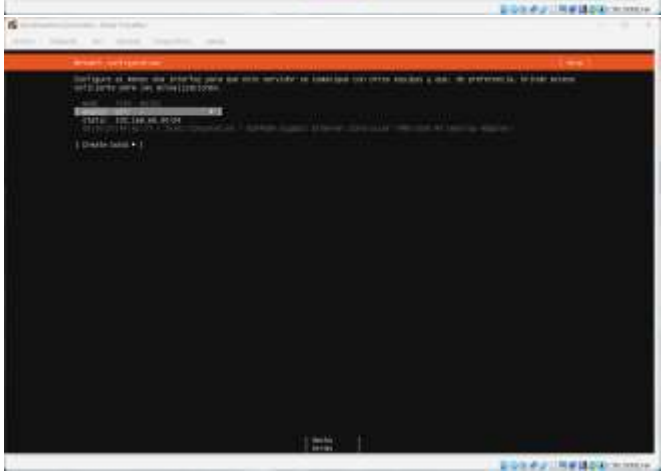


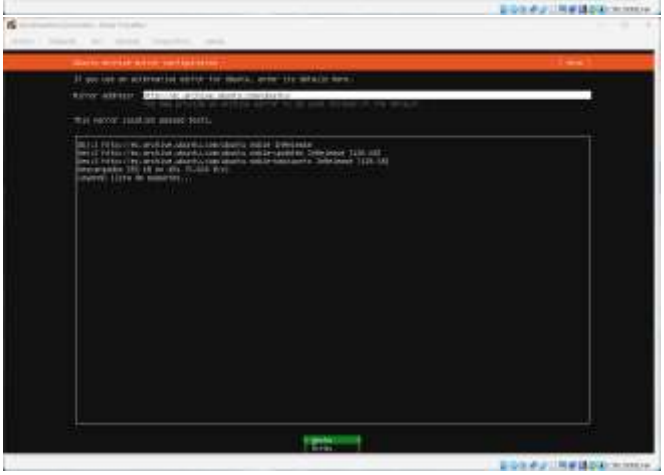


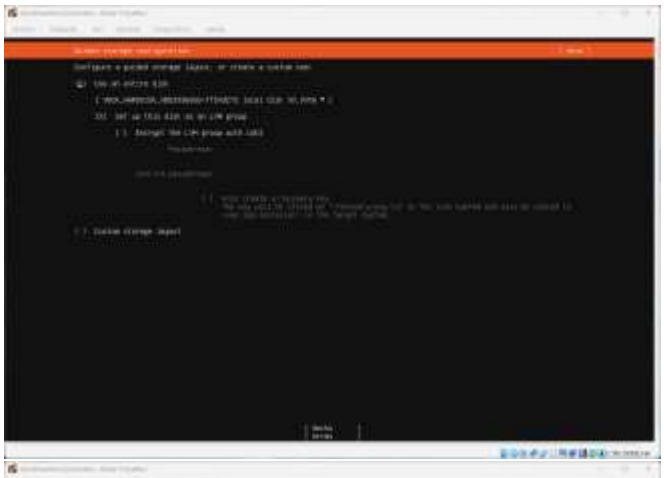


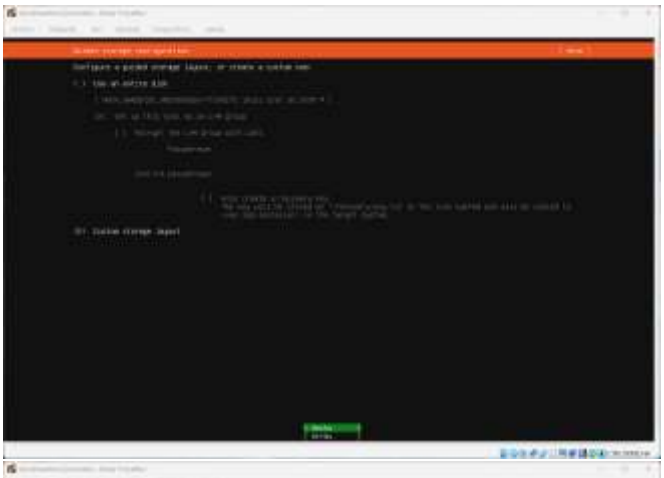


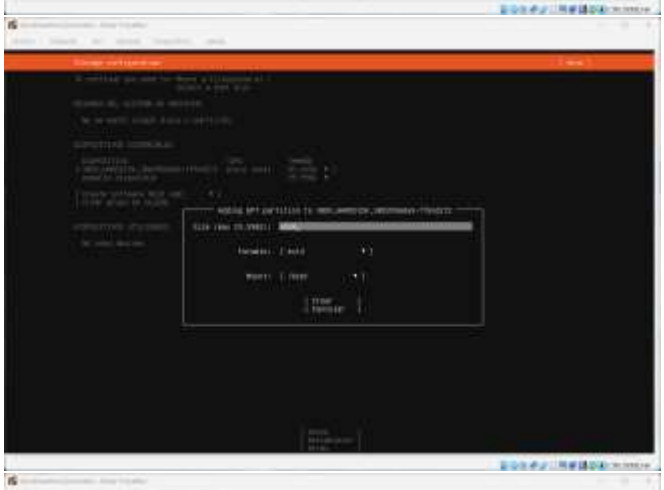


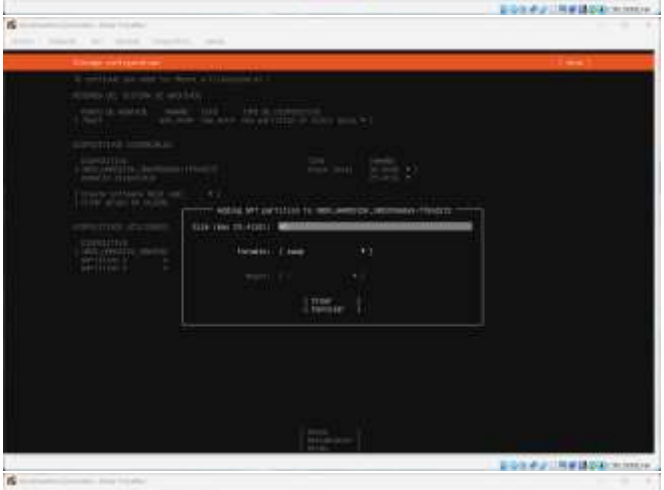


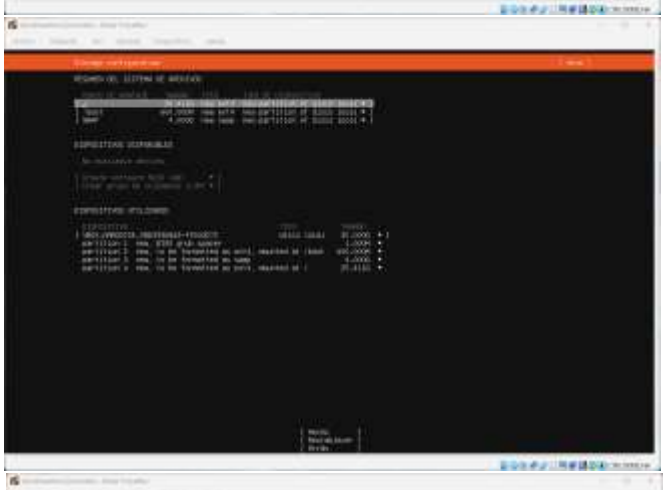


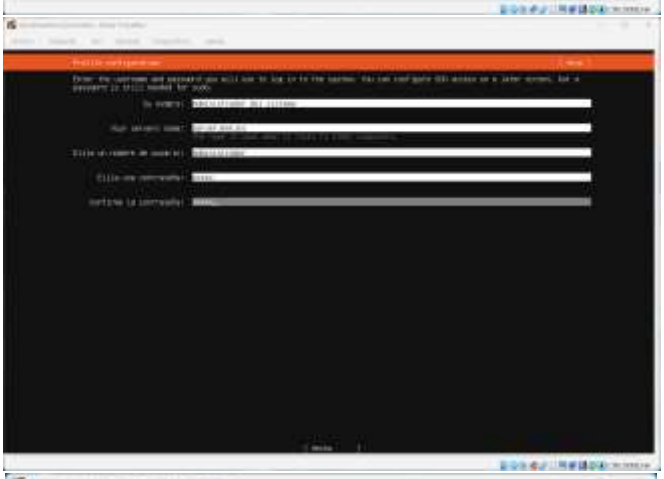


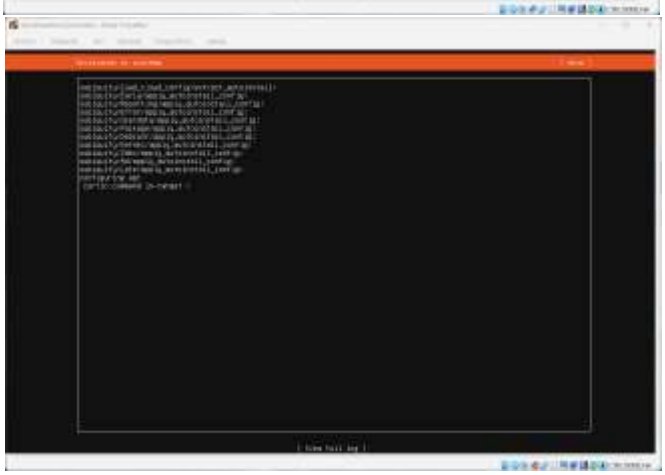


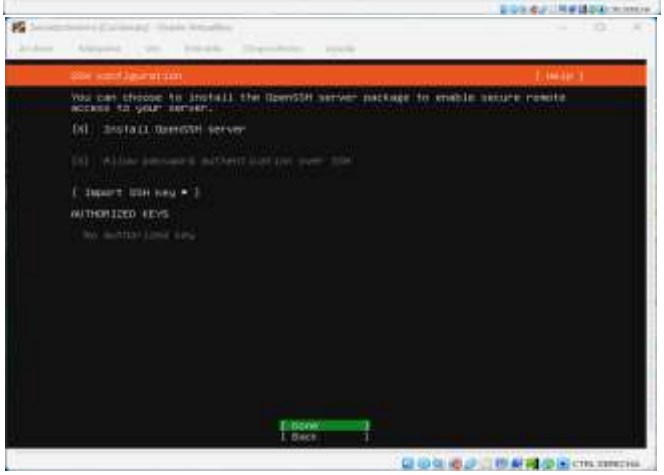


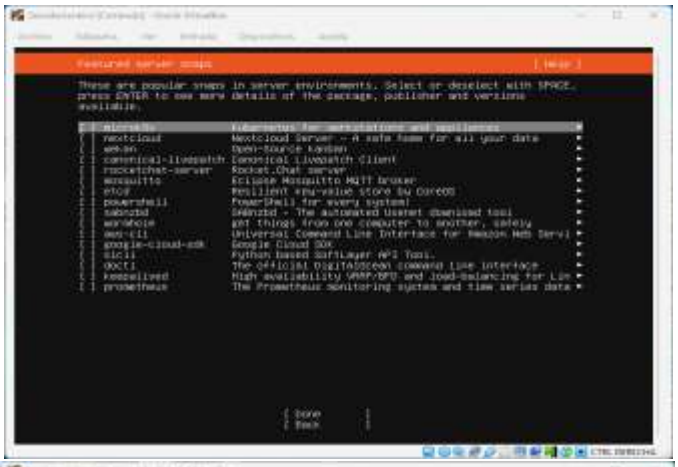


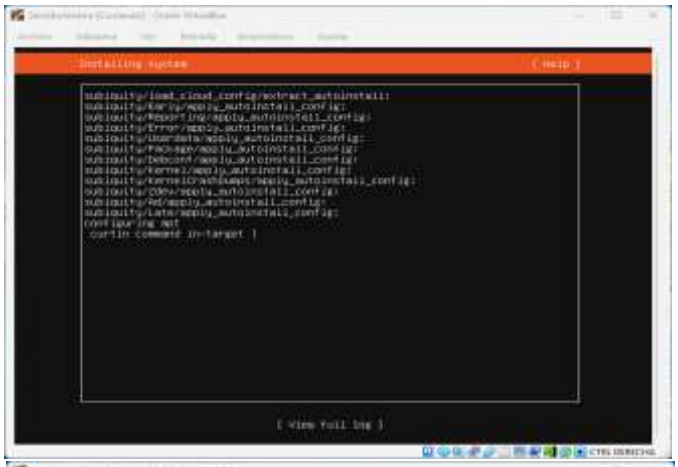


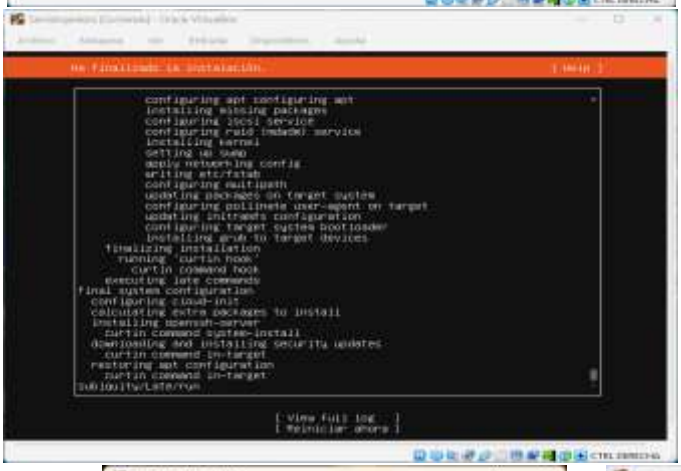


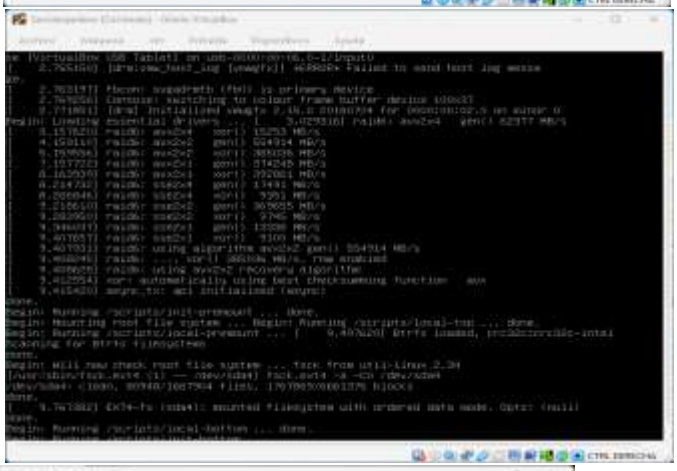


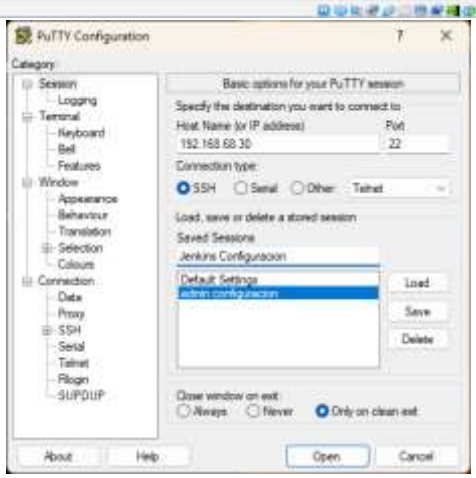

SSH configuration
[Help]
You can choose to install the OpenSSH server package to enable secure remote access to your server.
(X) Install OpenSSH server
[Import SSH key]
AUTHORIZED KEYS
[Done]
[Back]

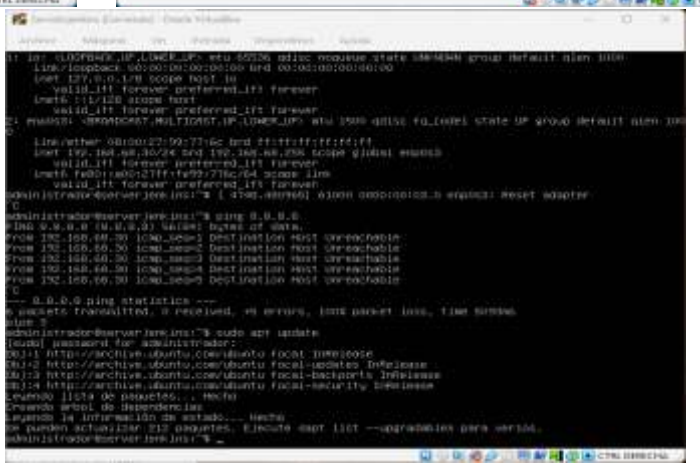


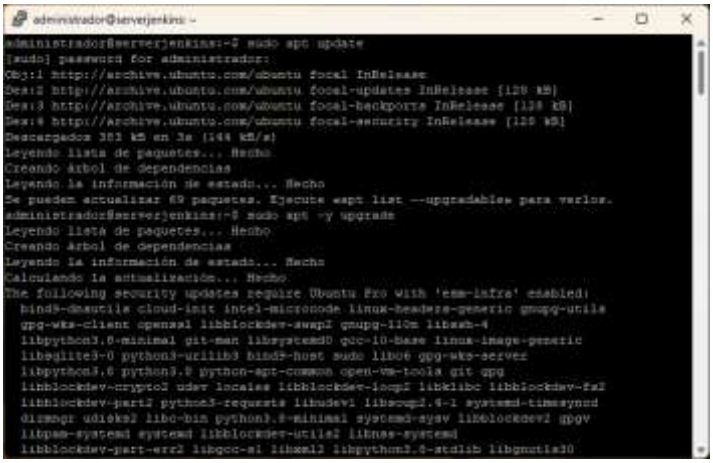


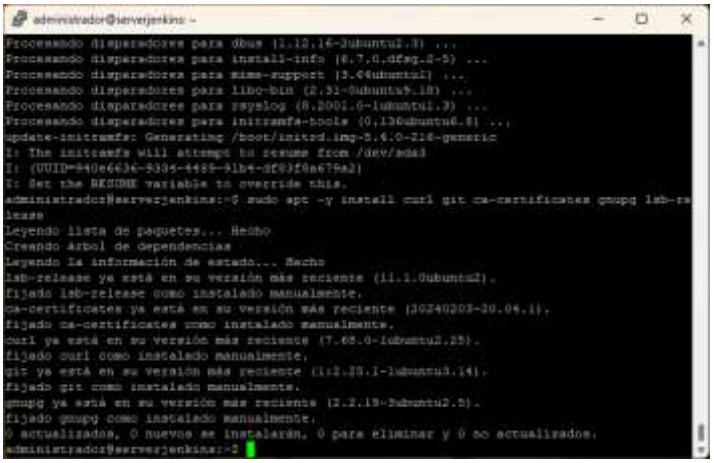




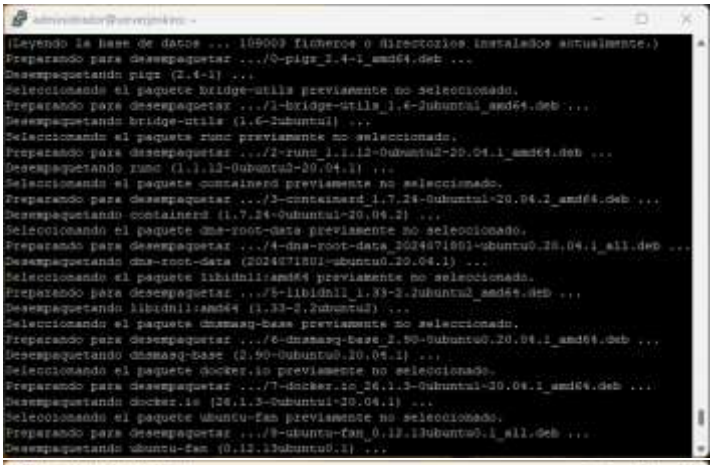

PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
192.168.68.30
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Jenkins Configuracion
Default Settings
admin configuracion
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel

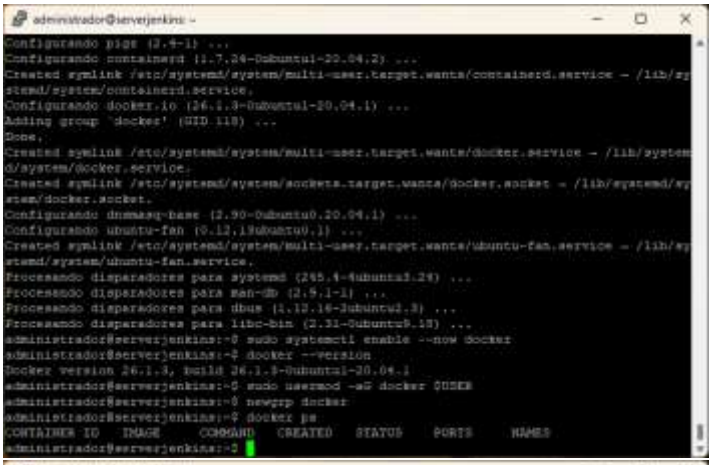


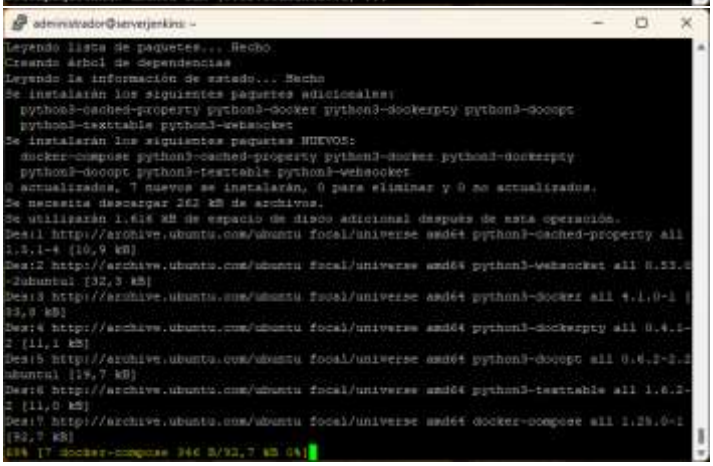


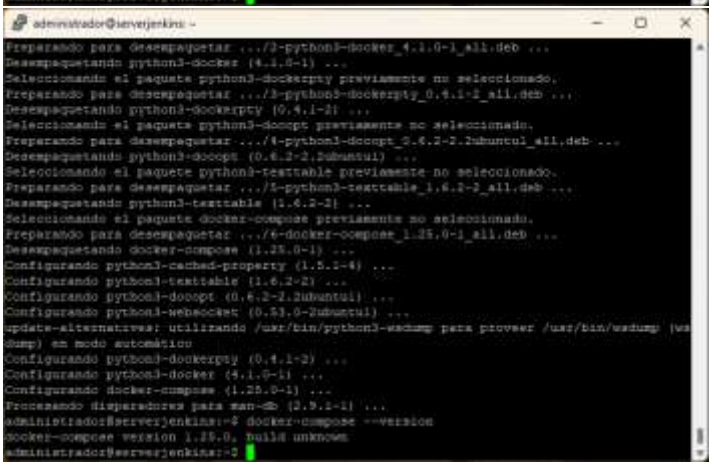


2. INSTALACION DOCKER Y DOCKER COMPOSE

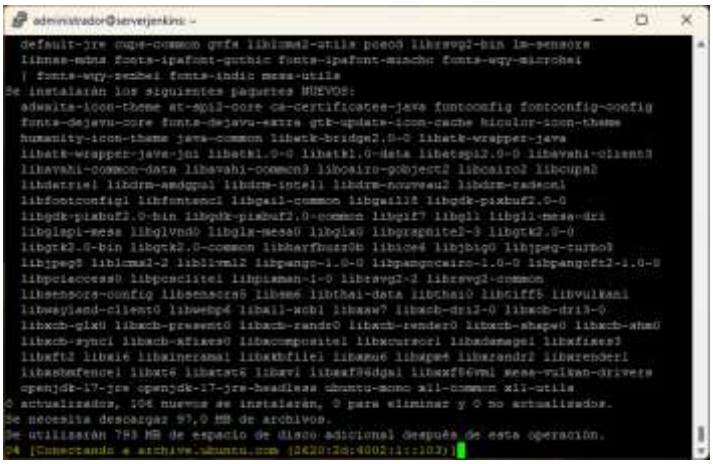


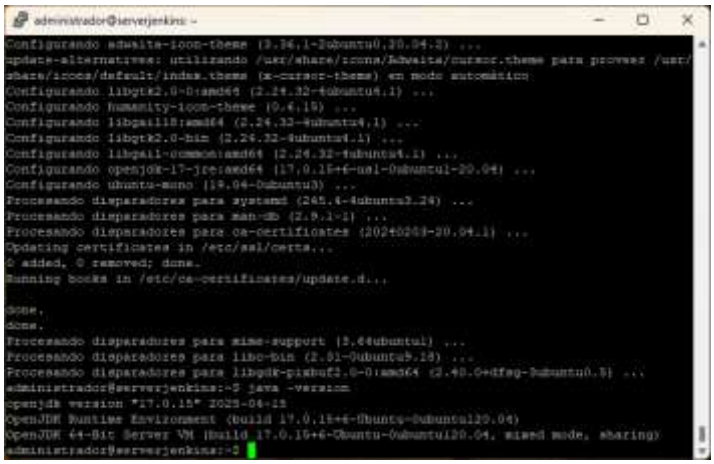


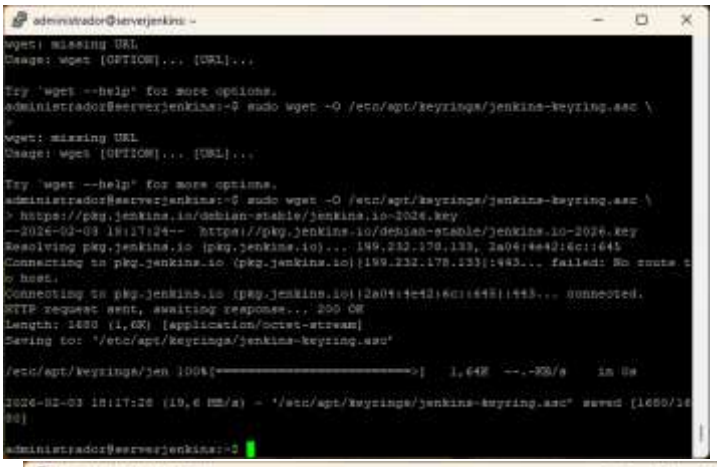


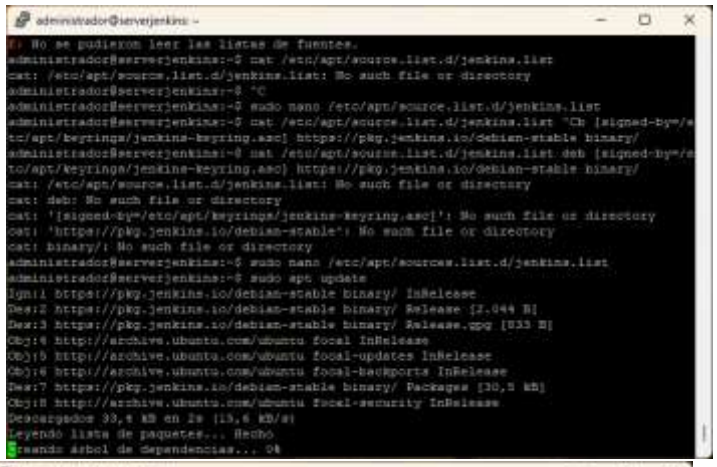


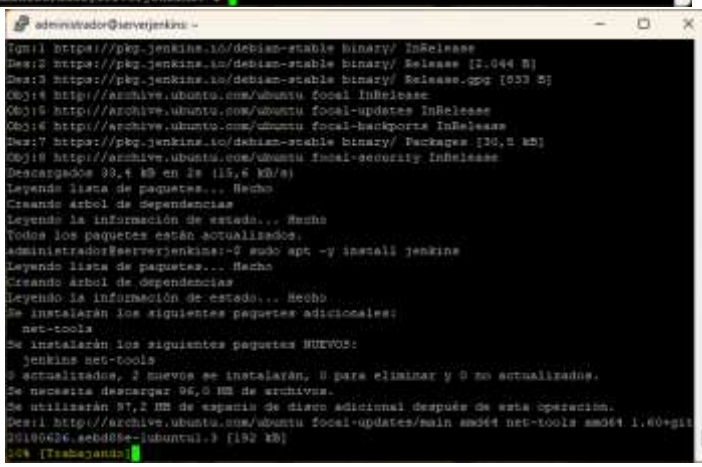
3. INSTALACION DE JENKINS EN LA VM

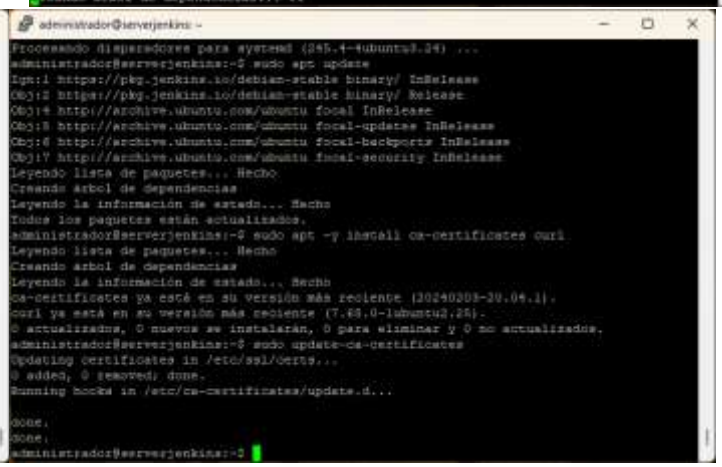


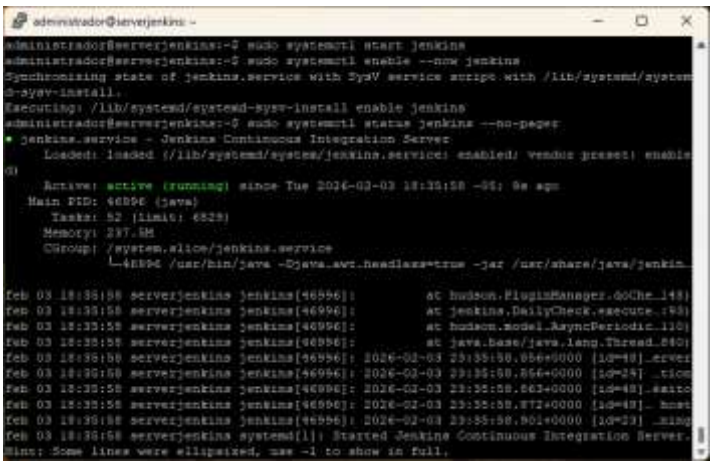



administrador@serverjenkins: ~
wget: missing URL
Usage: wget [OPTION]... [URL]...
Try 'wget --help' for more options.
administrador@serverjenkins:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
wget: missing URL
Usage: wget [OPTION]... [URL]...
Try 'wget --help' for more options.
administrador@serverjenkins:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
> https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-02-03 18:17:24-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 199.232.178.133, 2a04:4e42:6c::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|199.232.178.133|:443... failed: No route t
o host.
Connecting to pkg.jenkins.io (pkg.jenkins.io)|2a04:4e42:6c::645|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1,6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'
/etc/apt/keyrings/jen 100%[===========>] 1,64K --.-KB/s in 0s
2026-02-03 18:17:26 (19,6 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/16
80]
administrador@serverjenkins:~$


administrador@serverjenkins: ~
E: No se pudieron leer las listas de fuentes.
administrador@serverjenkins:~$ cat /etc/apt/source.list.d/jenkins.list
cat: /etc/apt/source.list.d/jenkins.list: No such file or directory
administrador@serverjenkins:~$ ^C
administrador@serverjenkins:~$ sudo nano /etc/apt/source.list.d/jenkins.list
administrador@serverjenkins:~$ cat /etc/apt/source.list.d/jenkins.list "Ch [signed-by=/e
tc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
administrador@serverjenkins:~$ cat /etc/apt/source.list.d/jenkins.list deb [signed-by=/e
tc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
cat: /etc/apt/source.list.d/jenkins.list: No such file or directory
cat: deb: No such file or directory
cat: '[signed-by=/etc/apt/keyrings/jenkins-keyring.asc]': No such file or directory
cat: 'https://pkg.jenkins.io/debian-stable': No such file or directory
cat: binary/: No such file or directory
administrador@serverjenkins:~$ sudo nano /etc/apt/source.list.d/jenkins.list
administrador@serverjenkins:~$ sudo apt update
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Des:2 https://pkg.jenkins.io/debian-stable binary/ Release [2.044 B]
Des:3 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Obj:4 http://archive.ubuntu.com/ubuntu focal InRelease
Obj:5 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Obj:6 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Des:7 https://pkg.jenkins.io/debian-stable binary/ Packages [30,5 kB]
Obj:8 http://archive.ubuntu.com/ubuntu focal-security InRelease
Descargados 93,4 kB en 2s (15,6 kB/s)
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... 0%


administrador@serverjenkins: ~
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Des:2 https://pkg.jenkins.io/debian-stable binary/ Release [2.044 B]
Des:3 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Obj:4 http://archive.ubuntu.com/ubuntu focal InRelease
Obj:5 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Obj:6 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Des:7 https://pkg.jenkins.io/debian-stable binary/ Packages [30,5 kB]
Obj:8 http://archive.ubuntu.com/ubuntu focal-security InRelease
Descargados 93,4 kB en 2s (15,6 kB/s)
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Todos los paquetes están actualizados.
administrador@serverjenkins:~$ sudo apt -y install jenkins
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
net-tools
Se instalarán los siguientes paquetes NUEVOS:
jenkins net-tools
0 actualizados, 2 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
Se necesita descargar 96,0 MB de archivos.
Se utilizarán 97,2 MB de espacio de disco adicional después de esta operación.
Des:1 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 net-tools amd64 1.60+git
20180626.aebd88e-1ubuntu1.3 [192 kB]
10% [Trabajando]


administrador@serverjenkins: ~
Procesando disparadores para systemd (245.4-4ubuntu3.24) ...
administrador@serverjenkins:~$ sudo apt update
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Obj:2 https://pkg.jenkins.io/debian-stable binary/ Release
Obj:4 http://archive.ubuntu.com/ubuntu focal InRelease
Obj:5 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Obj:6 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Obj:7 http://archive.ubuntu.com/ubuntu focal-security InRelease
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Todos los paquetes están actualizados.
administrador@serverjenkins:~$ sudo apt -y install ca-certificates curl
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
ca-certificates ya está en su versión más reciente (20240203~20.04.1).
curl ya está en su versión más reciente (7.68.0-1ubuntu2.25).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 0 no actualizados.
administrador@serverjenkins:~$ sudo update-ca-certificates
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
done.
administrador@serverjenkins:~$


administrador@serverjenkins: ~
administrador@serverjenkins:~$ sudo systemctl start jenkins
administrador@serverjenkins:~$ sudo systemctl enable --now jenkins
Synchronizing state of jenkins.service with SysV service script with /lib/systemd/system
d-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable jenkins
administrador@serverjenkins:~$ sudo systemctl status jenkins --no-pager
● jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enable
d)
Active: active (running) since Tue 2026-02-03 18:35:58 -05; 9s ago
Main PID: 46896 (java)
Tasks: 52 (limit: 6829)
Memory: 297.5M
CGroup: /system.slice/jenkins.service
└─46896 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkin
feb 03 18:35:58 serverjenkins jenkins[46996]: at hudson.PluginManager.doChe
feb 03 18:35:58 serverjenkins jenkins[46996]: at jenkins.DailyCheck.execute
feb 03 18:35:58 serverjenkins jenkins[46996]: at hudson.model.AsyncPeriodic
feb 03 18:35:58 serverjenkins jenkins[46996]: at java.base/java.lang.Thread
feb 03 18:35:58 serverjenkins jenkins[46996]: 2026-02-03 23:35:58.056+0000 [id=48]
feb 03 18:35:58 serverjenkins jenkins[46996]: 2026-02-03 23:35:58.056+0000 [id=29]
feb 03 18:35:58 serverjenkins jenkins[46996]: 2026-02-03 23:35:58.063+0000 [id=48]
feb 03 18:35:58 serverjenkins jenkins[46996]: 2026-02-03 23:35:58.072+0000 [id=48]
feb 03 18:35:58 serverjenkins jenkins[46996]: 2026-02-03 23:35:58.901+0000 [id=23]
feb 03 18:35:58 serverjenkins systemd[1]: Started Jenkins Continuous Integration Server.
Hint: Some lines were ellipsized, use -l to show in full.

Empezando
Desbloquear Jenkins
Para garantizar que el administrador configure Jenkins de forma segura, se ha escrito una contraseña en el registro (¿no está seguro de dónde encontrarla?) y en este archivo en el servidor:
Copie la contraseña de cualquiera de las ubicaciones y péguela a continuación.
Contraseña de administrador
Continuar

Empezando
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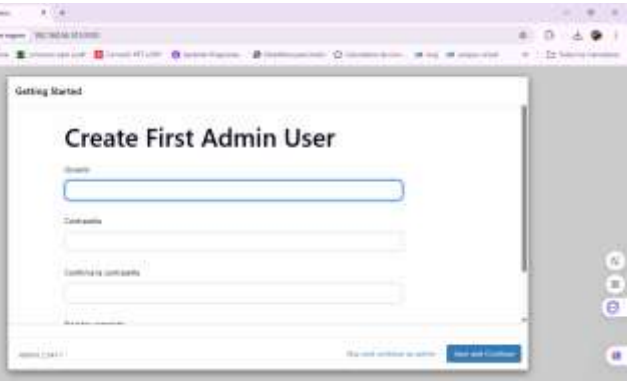
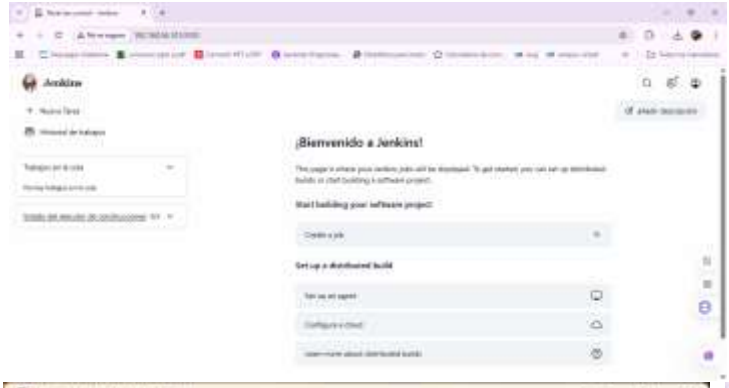
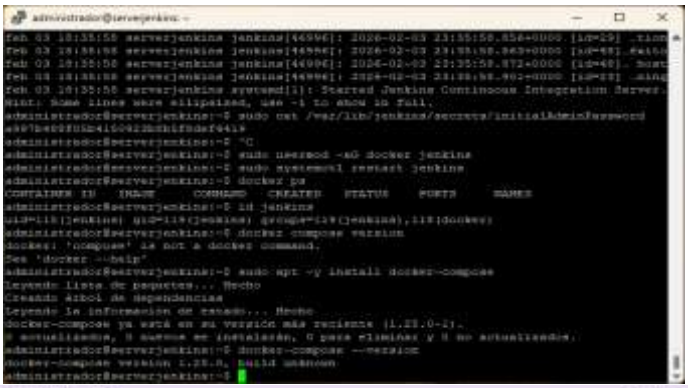
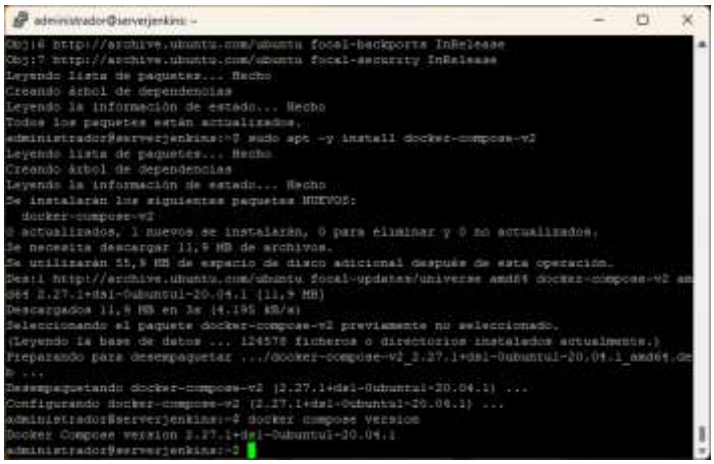
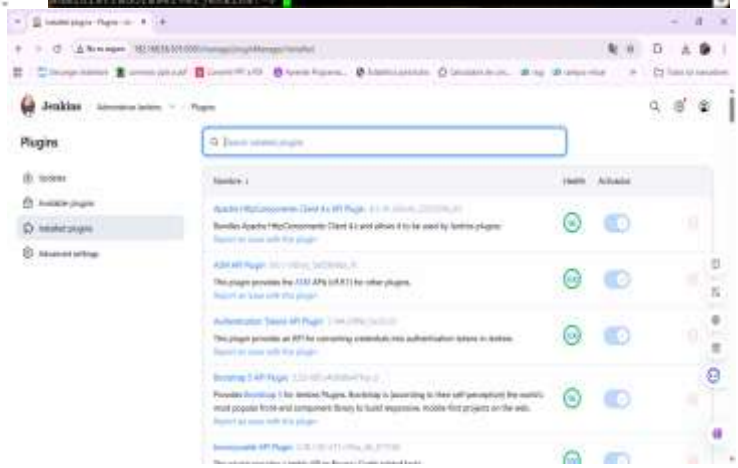
Continuar


Getting Started
Create First Admin User
Usuario
Contraseña
Confirmar contraseña
Nombre completo
Save and Continue

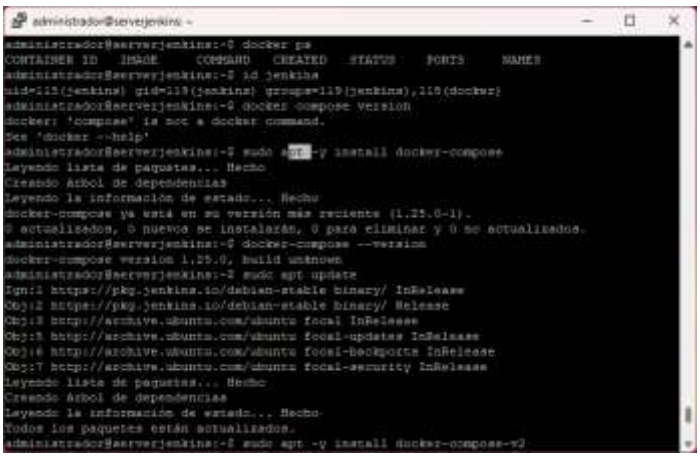




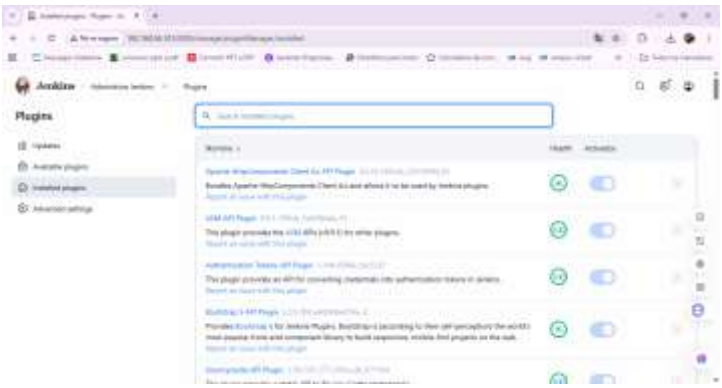




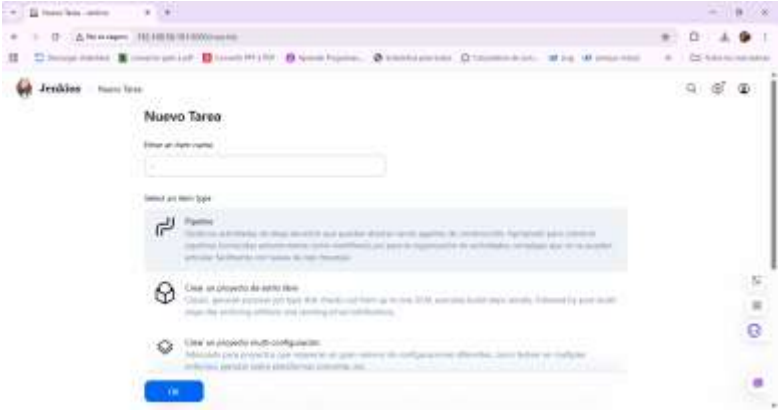
4. CONFIGURAR JENKINS PARA USAR DOCKER



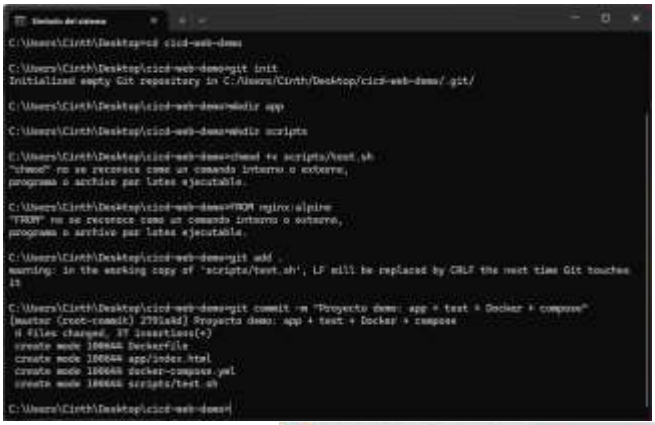
5. INSTALACION DE PUGLINS MINIMOS

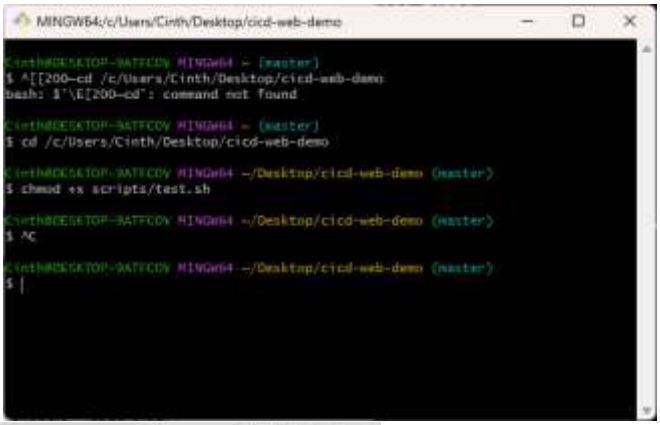


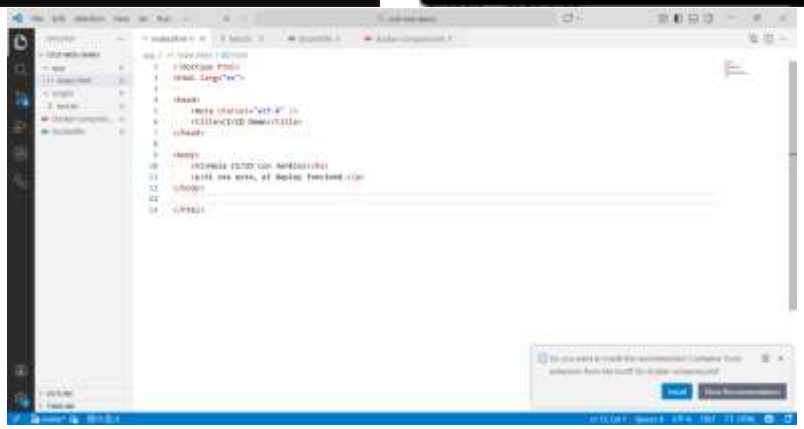
6. VERIFICAR PIPELINE ESTE DISPONIBLE



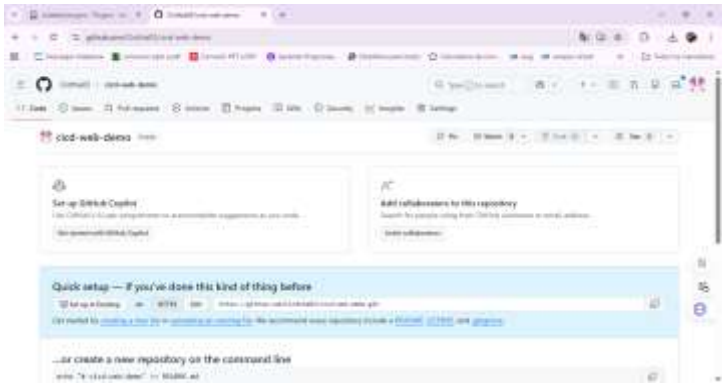
7. CREAR PROYECTO DE EJEMPLO

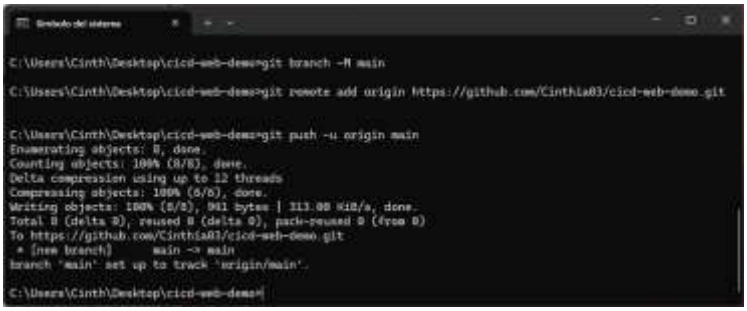


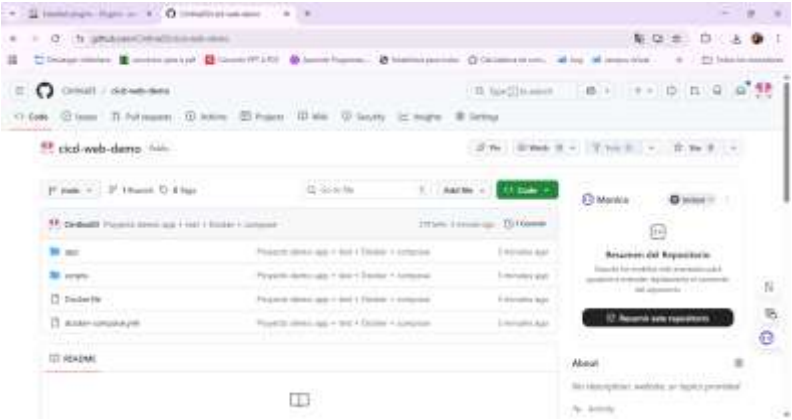




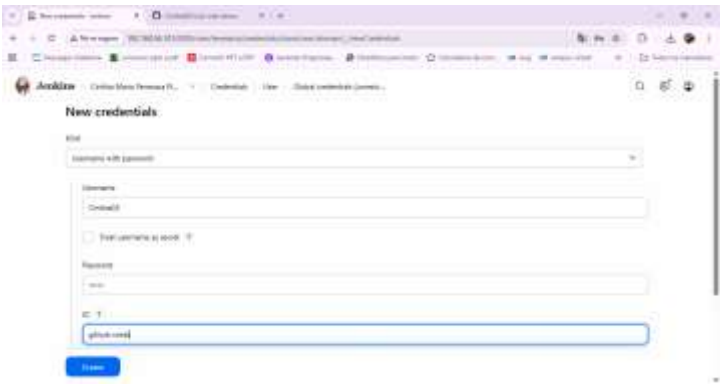
8. SUBIR REPOSITORIO A GITHUB

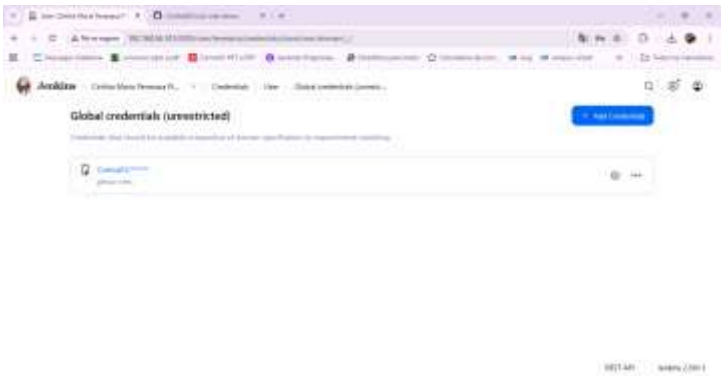




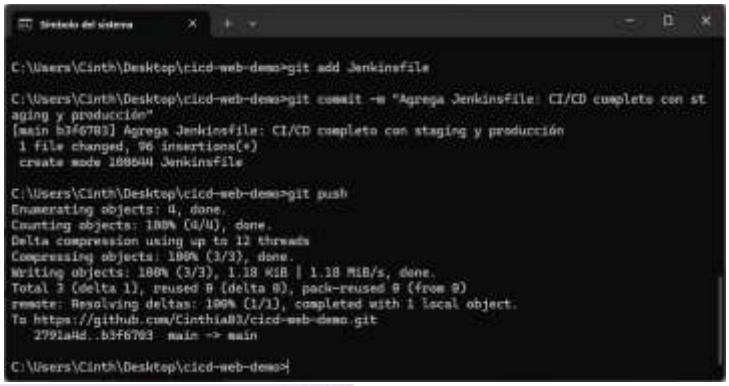


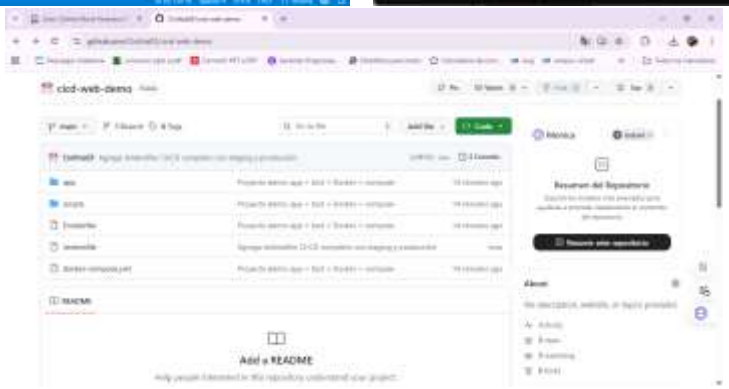
9. CREAR CREDENCIALES EN JENKINS PARA GITHUB



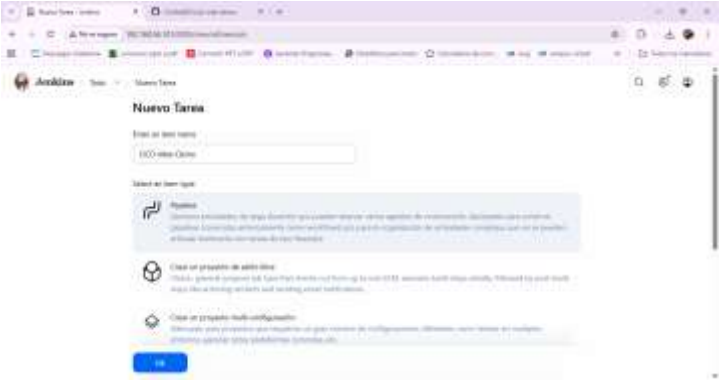


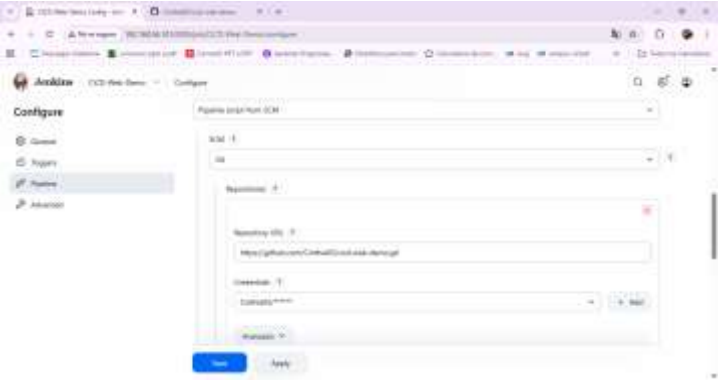
10. CREAR PIPELINE CI/CD EN JENKINS DESDE JENKINSFILE

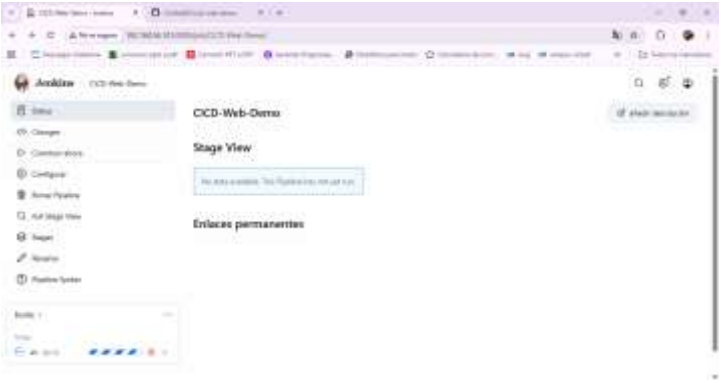


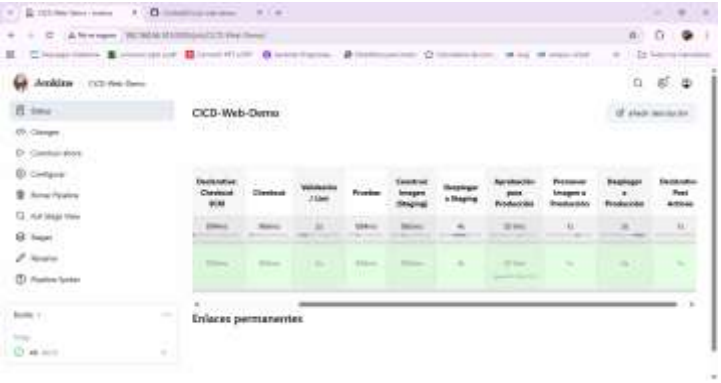


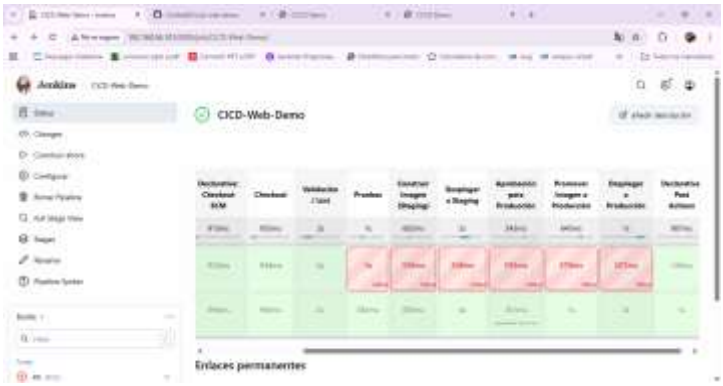
11. CONFIGURAR EL JOB PIPELINE EN JENKINS









12. VERFICACION (STAGING Y PRODUCCION)

13. ACTIVIDADES GUIADAS PARA DEMOSTRAR CI/CD